

Quadratic Equations – Complete Extraction

5.2 (remaining), 5.3 Nature of Roots & 5.4 Relations between Roots

Contents

1	5.2 Equations Reducible to Quadratic Form (Continued)	2
1.1	Exercise 5.2 – Questions 8 to 18	2
1.2	Answers – Exercise 5.2 (Questions 8–18)	2
2	5.3 Nature of Roots of a Quadratic Equation	3
2.1	Illustrative Examples	3
2.2	Answers – Illustrative Examples (5.3)	4
2.3	Exercise 5.3	4
2.4	Answers – Exercise 5.3 (Final Answers Only)	5
3	5.4 Relations between Roots and Coefficients	6
3.1	Illustrative Examples	6
3.2	Answers – Illustrative Examples (5.4) – Final Answers Only	7

1 5.2 Equations Reducible to Quadratic Form (Continued)

1.1 Exercise 5.2 – Questions 8 to 18

Solve the following equations:

8. $x(x+2)(x^2-1) = -1$

9. $(x+1)(2x+3)(2x+5)(x+3) = 945$

10. $(x^2+x-6)(x^2-3x-4) = 24$

11. (i) $\sqrt{\frac{x}{1-x}} + \sqrt{\frac{1-x}{x}} = \frac{13}{6}$

12. (ii) $\sqrt{\frac{4x-1}{4x+1}} - \sqrt{\frac{4x+1}{4x-1}} = \frac{8}{3}, x \neq \pm\frac{1}{4}$

13. (i) $\sqrt{3x+4} = x$ (ii) $\sqrt{3x+1} - \sqrt{x-1} = 2$

14. (i) $\sqrt{x+1} + \sqrt{x-1} = 1$ (ii) $\sqrt{x+3} + \sqrt{x-3} = 6$

15. (i) $\sqrt{3x^2-4x+34} - \sqrt{3x^2-4x-11} = 5$

16. (ii) $\sqrt{3x^2-7x-30} - \sqrt{2x^2-7x-5} = x-5$

17. $12 + 9\sqrt{(x-1)(3x+2)} = 3x^2 - x$

18. $\frac{\sqrt{x^2+1} - \sqrt{x^2-1}}{\sqrt{x^2+1} + \sqrt{x^2-1}} = \frac{1}{3}$

19. (i) $\left(x + \frac{1}{x}\right)^2 - 2\left(x - \frac{1}{x} + 4\right) = 11$

20. (ii) $12x^4 - 56x^3 + 89x^2 - 56x + 12 = 0$

21. Solve $(x^2+2)^2 + 8x^2 = 6x(x^2+2)$

1.2 Answers – Exercise 5.2 (Questions 8–18)

8. $-\frac{1+\sqrt{5}}{2}, -\frac{1-\sqrt{5}}{2}$

9. $2, -6, \frac{-4+i\sqrt{59}}{2}$

10. $0, 1, \frac{1+\sqrt{57}}{2}$

11. (i) $\frac{9}{13}, \frac{4}{13}$

12. (ii) $-\frac{5}{16}$

13. (i) 4 (ii) 1, 5

14. (i) No solution (ii) $\frac{37}{4}$

15. (i) $3, -\frac{5}{3}$ (ii) $5, 6$

16. $6, -\frac{17}{3}$

17. $\pm\sqrt{\frac{5}{3}}$

18. (i) $\frac{5 \pm \sqrt{29}}{2}, -\frac{3 \pm \sqrt{13}}{2}$

19. (ii) $\frac{1}{2}, \frac{2}{3}, \frac{3}{2}, 2$

20. $1 \pm i, 2 \pm \sqrt{2}$

2 5.3 Nature of Roots of a Quadratic Equation

2.1 Illustrative Examples

Example 1. Discuss the nature of the roots of the following equations:

(i) $4x^2 - 12x + 9 = 0$

(ii) $3x^2 - 10x + 3 = 0$

(iii) $9x^2 - 2 = 0$

(iv) $x^2 + x + 1 = 0$

Example 2. Discuss the nature of the roots of the following equations:

(i) $4x^2 - 4\sqrt{3}x + 3 = 0$

(ii) $2\sqrt{3}x^2 - 5x + \sqrt{3} = 0$

Example 3. If $a, b, h \in \mathbb{R}$, show that the equation $(x - a)(x - b) = h^2$ has real roots.

Example 4. If the roots of $ax^2 + x + b = 0$ are real and distinct, show that the roots of $\frac{x^2 + 1}{x} = 4\sqrt{ab}$ are imaginary.

Example 5. Given that $a, b, c \in \mathbb{Q}$, show that the roots of $(b - c)x^2 + (c - a)x + (a - b) = 0$ are rational. Find the condition for the roots to be equal.

Example 6. Discuss the nature of the roots of the equation $(m + 6)x^2 + (m + 6)x + 2 = 0$.

Example 7. For what value of k does the expression $x^2 - 2x(1 + 3k) + 7(3 + 2k)$ become a perfect square? Also write the expression as a perfect square.

Example 8. Determine a positive real value of k such that both the equations $x^2 + kx + 64 = 0$ and $x^2 - 8x + k = 0$ may have real roots.

Example 9. Show that if p, q, r, s are real numbers and $pr = 2(q + s)$, then at least one of the equations $x^2 + px + q = 0$ and $x^2 + rx + s = 0$ has real roots.

2.2 Answers – Illustrative Examples (5.3)

- Example 1. (i) Rational and equal
 (ii) Rational and unequal
 (iii) Irrational and unequal (conjugates)
 (iv) Pair of complex conjugates

- Example 2. (i) Real and equal
 (ii) Real and unequal

Example 3. Proven ($\Delta \geq 0$)

Example 4. Proven (second equation has negative discriminant)

Example 5. Roots are rational. Equal when $b = \frac{a+c}{2}$

- Example 6.
 - Equal when $m = 2$ or $m = -6$
 - Real & unequal when $m < -6$ or $m > 2$
 - Complex when $-6 < m < 2$

Example 7. $k = 2((x-7)^2)$ or $k = -\frac{10}{9}\left(x + \frac{7}{3}\right)^2$

Example 8. $k = 16$

Example 9. Proven

2.3 Exercise 5.3

1. Find the nature of roots (without solving):
 - (i) $x^2 + 9 = 0$
 - (ii) $4x^2 - 24x + 35 = 0$
 - (iii) $x^2 - 2\sqrt{2}x + 1 = 0$
 - (iv) $2x^2 - 2\sqrt{5}x + 3 = 0$
2. (i) Prove that roots of $x^2 - 2ax + a^2 - b^2 - c^2 = 0$ are always real.
3. (ii) Show that roots of $(x-a)(x-b) = abx^2$ are always real. When are they equal?
4. Show that roots of $(x-a)(x-b) + (x-b)(x-c) + (x-c)(x-a) = 0$ are always real. Find condition for equal roots.
5. Discuss nature of roots:
 - (i) $\sqrt{3}x^2 - 2x - \sqrt{3} = 0$
 - (ii) $x^2 - (p+1)x + p = 0$ ($p \in \mathbb{Q}$)
 - (iii) $(x-a)(x-b) = ab$ ($a, b \in \mathbb{R}$)
6. Find m so that roots of $(4+m)x^2 + (m+1)x + 1 = 0$ are equal.
7. Show that roots of $x^2 + 2(3a+5)x + 2(9a^2+25) = 0$ are complex unless $a = \frac{5}{3}$.

8. If $a, b, c, d \in \mathbb{R}$, show that roots of $(a^2 + c^2)x^2 + 2(ab + cd)x + (b^2 + d^2) = 0$ cannot be real unless equal.
9. If $a, b, c \in \mathbb{Q}$ and $a + b + c = 0$, show that roots of $ax^2 + bx + c = 0$ are rational. Is converse true?
10. Prove that $x^2 + px - 1 = 0$ has real and distinct roots for all real p .
11. Show that roots of $x^2 - 2\left(\frac{m+1}{m}\right)x + 3 = 0$ are real for all non-zero real m .
12. If roots of $ax^2 + 2cx + b = 0$ are real and distinct, show that roots of $x^2 - 2(a+b)x + a^2 + b^2 + 2c^2 = 0$ are non-real complex.
13. If roots of $a(b-c)x^2 + b(c-a)x + c(a-b) = 0$ are equal, show that $\frac{2}{b} = \frac{1}{a} + \frac{1}{c}$.
14. For what value of k is $(4-k)x^2 + 2(k+2)x + (8k+1)$ a perfect square?
15. If roots of $x^2 - 8x + m(m-6) = 0$ are real and distinct, find all possible values of m .

2.4 Answers – Exercise 5.3 (Final Answers Only)

1. (i) Complex conjugates (ii) Rational & unequal (iii) Real & unequal (iv) Complex conjugates
2. (ii) Equal when $a = b = 0$
3. Equal when $a = b = c$; roots = a, a
4. (i) Real & distinct (ii) Rational & distinct ($p \neq 1$); equal when $p = 1$ (iii) Real & distinct ($a + b \neq 0$); both zero when $a + b = 0$
5. $m = 5$ or $m = -3$
6. Proven
7. Proven
8. Proven (converse false)
9. Proven
10. Proven
11. Proven
12. Proven
13. $k = 0$ or $k = 3$
14. $-2 < m < 8$

3 5.4 Relations between Roots and Coefficients

3.1 Illustrative Examples

Example 1. If α, β are roots of $x^2 - 4x + 2 = 0$, find:

- (i) $\alpha^2 + \beta^2$
- (ii) $\alpha^2 - \beta^2$
- (iii) $\alpha^3 + \beta^3$
- (iv) $\frac{1}{\alpha} + \frac{1}{\beta}$

Example 2. If α, β are roots of $ax^2 + bx + c = 0$, show that $ax^2 + bx + c = a(x - \alpha)(x - \beta)$.

Example 3. For $(k - 1)x^2 = kx - 1$ ($k \neq 1$), find k so that:

- (i) one root is -3
- (ii) sum of roots $= 2$
- (iii) product of roots $= -3$
- (iv) roots are equal
- (v) roots numerically equal but opposite in sign

Example 4. If one root of $3x^2 - 7px + 10p = 0$ is 5, find p and the other root.

Example 5. The sum of roots of $\frac{1}{x+a} + \frac{1}{x+b} = \frac{1}{c}$ is zero. Prove that product of roots is $-\frac{1}{2}(a^2 + b^2)$.

Example 6. If α, β are roots of $ax^2 + bx + c = 0$, evaluate:

- (i) $\alpha^3\beta + \alpha\beta^3$
- (ii) $\frac{\alpha^2}{\beta} + \frac{\beta^2}{\alpha}$
- (iii) $(a\alpha + b)^2 + (a\beta + b)^2$

Example 7. Form an equation whose roots are:

- (i) $2, -\frac{1}{2}$
- (ii) one root $\frac{1}{3+2\sqrt{2}}$ (rational coefficients)
- (iii) one root $-2 + i$ (real coefficients)

Example 8. If α, β are roots of $ax^2 + bx + c = 0$, form equation with roots:

- (i) $-\alpha, -\beta$
- (ii) $\frac{1}{\alpha}, \frac{1}{\beta}$
- (iii) α^2, β^2
- (iv) α^3, β^3

Example 9. Find the equation whose roots are n times the roots of $ax^2 + bx + c = 0$.

- Example 10. Find p such that difference of roots of $x^2 - px + 10 = 0$ is 3. Also form equation whose roots are $(\alpha + \beta)^2$ and $\alpha^2\beta - \alpha\beta^2$.
- Example 11. Let p, q be roots of $3x^2 + 6x + 2 = 0$. Form equation with roots $-\frac{p^2}{q}, -\frac{q^2}{p}$.
- Example 12. (i) If roots of $x^2 - lx + m = 0$ differ by 1, prove $l^2 = 4m + 1$.
- Example 13. (ii) Find condition that roots of $ax^2 + bx + c = 0$ are in ratio $p : q$.
- Example 14. If α, β are roots of $x^2 + ax + b = 0$, prove that $\frac{\alpha}{\beta}$ is a root of $bx^2 + (2b - a^2)x + b = 0$.
- Example 15. If roots of $ax^2 + bx + b = 0$ ($b^2 \geq 4ab$) are in ratio $p : q$, prove $\sqrt{\frac{p}{q}} + \sqrt{\frac{q}{p}} = \sqrt{\frac{b}{a}}$.
- Example 16. Roots of $px^2 - 2(p+2)x + 3p = 0$ differ by 2. Find p and the roots.
- Example 17. Solve $x^2 + px + 45 = 0$ given that square of difference of roots is 144.
- Example 18. If difference of roots of $x^2 + px + q = 0$ equals difference of roots of $x^2 + qx + p = 0$, prove $p + q + 4 = 0$ ($p \neq q$).
- Example 19. (i) If ratio of roots of $x^2 + px + q = 0$ equals ratio of roots of $x^2 + bx + c = 0$, prove $p^2c = b^2q$.
- Example 20. (ii) If roots α, β and γ, δ are in G.P., prove same.
- Example 21. Coefficient of x in $x^2 + px + q = 0$ was taken as 17 instead of 13; roots found as -2 and -15. Find roots of original equation.
- Example 22. If one root of $ax^2 + bx + c = 0$ is square of the other, prove $b^3 + ac(c + a) = 3abc$.
- Example 23. If one root of $ax^2 + bx + c = 0$ is n -th power of the other, show that $(ac^n)^{n+1} + (a^n c)^{n+1} + b = 0$.
- Example 24. If $x^2 + ax + b = 0$ and $x^2 + bx + a = 0$ have a common root and $a \neq b$, prove $a + b + 1 = 0$.
- Example 25. If $3x^2 + px + 1 = 0$ and $2x^2 + qx + 1 = 0$ have a common root, show that $2p^2 + 3q^2 - 5pq + 1 = 0$.

3.2 Answers – Illustrative Examples (5.4) – Final Answers Only

Example 1. (i) 12 (ii) $\pm 8\sqrt{2}$ (iii) 40 (iv) 2

Example 2. Proven

Example 3. (i) $\frac{2}{3}$ (ii) 2 (iii) $\frac{2}{3}$ (iv) 2 (v) 0

Example 4. $p = 3$, other root = 2

Example 5. Proven

Example 6. (i) $\frac{c(b^2 - 2ac)}{a^3}$ (ii) $\frac{b(b^2 - 2ac)}{a^2c}$ (iii) $\frac{b^2(b^2 - 2ac)}{a^2}$

Example 7. (i) $2x^2 - 3x - 2 = 0$ (ii) $x^2 - 6x + 1 = 0$ (iii) $x^2 + 4x + 5 = 0$

Example 8. (i) $ax^2 - bx + c = 0$ (ii) $cx^2 + bx + a = 0$ (iii) $a^2x^2 - (b^2 - 2ac)x + c^2 = 0$ (iv) $a^3x^2 - (3abc - b^3)x + c^3 = 0$

Example 9. $ax^2 + nbx + n^2c = 0$

Example 10. $p = 7$ or -7 ; new equation $x^2 - 79x + 1470 = 0$

Example 11. $3x^2 - 18x + 2 = 0$

Example 12. (i) Proven (ii) $pqb^2 = ca(p + q)^2$

Example 13. Proven

Example 14. Proven

Example 15. $p = 2$ (roots 1, 3) or $p = -\frac{2}{3}$ (roots -3, -1)

Example 16. $p = 18$ (roots -3, -15) or $p = -18$ (roots 3, 15)

Example 17. Proven

Example 18. (i) & (ii) Proven ($p^2c = b^2q$)

Example 19. Original equation: $x^2 + 13x + 30 = 0$ (roots -3, -10)

Example 20. Proven

Example 21. Proven

Example 22. Proven

Example 23. Proven

Exercise 5.4

- If α, β are roots of the equation $x^2 - 4x + 2 = 0$, find the values of:
 - $\alpha^2 + \beta^2$
 - $\alpha^2 - \beta^2$
 - $\alpha^3 + \beta^3$
 - $\frac{1}{\alpha} + \frac{1}{\beta}$
- If α, β are the roots of the equation $ax^2 + bx + c = 0$, then show that $ax^2 + bx + c = a(x - \alpha)(x - \beta)$.
- For the quadratic equation $(k - 1)x^2 = kx - 1$ ($k \neq 1$), find k so that:
 - one root is -3
 - the sum of the roots is 2
 - the product of the roots is -3
 - the roots are equal
 - the roots are numerically equal but opposite in sign
- If one of the roots of $3x^2 - 7px + 10p = 0$ is 5, find p and also the other root.

5. The sum of the roots of the equation $\frac{1}{x+a} + \frac{1}{x+b} = \frac{1}{c}$ is zero. Prove that the product of the roots is $-\frac{1}{2}(a^2 + b^2)$.
6. If α, β are the roots of the equation $ax^2 + bx + c = 0$, evaluate:
- $\alpha^3\beta + \alpha\beta^3$
 - $\frac{\alpha^2}{\beta} + \frac{\beta^2}{\alpha}$
 - $(a\alpha + b)^2 + (a\beta + b)^2$
7. Form an equation whose roots are:
- $2, -\frac{1}{2}$
 - one root $\frac{1}{3+2\sqrt{2}}$ (with rational coefficients)
 - one root $-2 + i$ (with real coefficients)
8. If α, β are roots of the quadratic equation $ax^2 + bx + c = 0$, form an equation whose roots are:
- $-\alpha, -\beta$
 - $\frac{1}{\alpha}, \frac{1}{\beta}$
 - α^2, β^2
 - α^3, β^3
9. Find the equation whose roots are n times the roots of the equation $ax^2 + bx + c = 0$.
10. Find the value of p such that the difference of the roots of the equation $x^2 - px + 10 = 0$ is 3. If the roots of $x^2 - px + 10 = 0$ are α and β , find the quadratic equation whose roots are $(\alpha + \beta)^2$ and $\alpha^2\beta - \alpha\beta^2$.
11. Let p, q be roots of $3x^2 + 6x + 2 = 0$. Form an equation whose roots are $-\frac{p^2}{q}, -\frac{q^2}{p}$.
12. (i) If the roots of the equation $x^2 - lx + m = 0$ differ by 1, then prove that $l^2 = 4m + 1$.
13. (ii) Find the condition that roots of the equation $ax^2 + bx + c = 0$ may be in the ratio $p : q$.
14. If α, β are the roots of $x^2 + ax + b = 0$, then prove that $\frac{\alpha}{\beta}$ is a root of the equation $bx^2 + (2b - a^2)x + b = 0$.
15. If the roots of the equation $ax^2 + bx + b = 0$ ($b^2 \geq 4ab$) are in ratio $p : q$, prove that $\sqrt{\frac{p}{q}} + \sqrt{\frac{q}{p}} = \sqrt{\frac{b}{a}}$.
16. The roots of the equation $px^2 - 2(p+2)x + 3p = 0$ differ by 2. Find p and also the roots.
17. Solve the equation $x^2 + px + 45 = 0$, given that the square of the difference of its roots is 144.

18. If the difference of the roots of the equation $x^2 + px + q = 0$ be the same as that of the roots of $x^2 + qx + p = 0$, then prove that $p + q + 4 = 0$ ($p \neq q$).
19. (i) If the ratio of the roots of the equation $x^2 + px + q = 0$ is equal to the ratio of the roots of the equation $x^2 + bx + c = 0$, then prove that $p^2c = b^2q$.
20. (ii) If α, β are roots of $x^2 + px + q = 0$ and γ, δ are roots of $x^2 + bx + c = 0$ such that $\alpha, \beta, \gamma, \delta$ are in G.P., then prove that $p^2c = b^2q$.
21. The coefficient of x in the equation $x^2 + px + q = 0$ was taken as 17 in place of 13 and thus its roots were found to be -2 and -15 . Find the roots of the original equation.
22. If one root of the equation $ax^2 + bx + c = 0$ is the square of the other, prove that $b^3 + ac(c+a) = 3abc$.
23. If one root of the quadratic equation $ax^2 + bx + c = 0$ is equal to the n th power of the other root, then show that $(ac^n)^{n+1} + (a^n c)^{n+1} + b = 0$.
24. If the equations $x^2 + ax + b = 0$ and $x^2 + bx + a = 0$ have a common root and $a \neq b$, then prove that $a + b + 1 = 0$.
25. If the equations $3x^2 + px + 1 = 0$ and $2x^2 + qx + 1 = 0$ have a common root, show that $2p^2 + 3q^2 - 5pq + 1 = 0$.

Answers – Exercise 5.4 (Final Answers Only)

1. (i) 12 (ii) $\pm 8\sqrt{2}$ (iii) 40 (iv) 2
2. Proven
3. (i) $\frac{2}{3}$ (ii) 2 (iii) $\frac{2}{3}$ (iv) 2 (v) 0
4. $p = 3$, other root = 2
5. Proven
6. (i) $\frac{c(b^2 - 2ac)}{a^3}$ (ii) $\frac{b(b^2 - 2ac)}{a^2c}$ (iii) $\frac{b^2(b^2 - 2ac)}{a^2}$
7. (i) $2x^2 - 3x - 2 = 0$ (ii) $x^2 - 6x + 1 = 0$ (iii) $x^2 + 4x + 5 = 0$
8. (i) $ax^2 - bx + c = 0$ (ii) $cx^2 + bx + a = 0$ (iii) $a^2x^2 - (b^2 - 2ac)x + c^2 = 0$ (iv) $a^3x^2 - (3abc - b^3)x + c^3 = 0$
9. $ax^2 + nbx + n^2c = 0$
10. $p = 7$ or -7 ; new equation: $x^2 - 79x + 1470 = 0$
11. $3x^2 - 18x + 2 = 0$
12. (i) Proven (ii) $pqb^2 = ca(p + q)^2$
13. Proven
14. Proven
15. $p = 2$ (roots: 1, 3) or $p = -\frac{2}{3}$ (roots: $-3, -1$)

- 16. $p = 18$ (roots: $-3, -15$) or $p = -18$ (roots: $3, 15$)
- 17. Proven
- 18. (i) & (ii) Proven ($p^2c = b^2q$)
- 19. Original equation: $x^2 + 13x + 30 = 0$ (roots: $-3, -10$)
- 20. Proven
- 21. Proven
- 22. Proven
- 23. Proven